

the component used is correct or placed correctly. Apart from these component problems, there could be soldering-side problems as well. When solder is used in excess it could create a bridge between two nearby pins, while less solder creates a dry point. Manufacturing defect testers detect the analogue component type and its value, but can't test digital components. These aren't very costly either. One can use manufacturing defect analysers (MDAs), flying probe testers (FPTs), automated optical inspection (AOI) systems or automated X-ray inspection testers depending on the equipment's return on investment. Of these, manufacturing defect analysers are the cheapest.

In the same kind of environment, one can even use a high-end tester called in-circuit tester (ICT). An ICT is a full-capacity tester capable of testing analogue, digital and mixed-signal devices. It costs ten times more than an MDA. The rate of fault detection is very high using an ICT. So a manufacturer using an ICT charges more for its products. Besides doing what an MDA does, the ICT performs power-on testing as well. That is, when you apply the power, it checks individual components on the assembly for their functionality (whether they are performing as per their specifications or not). It, however, can't test the full product for its functionality.

ICTs are very generic. These are user-programmable. You can use an ICT to test different components by using a dedicated fixture and program written for the PCB. The fixture ensures electrical connection between the assembly and the tester. These fixtures and programs are very dedicated. That means if you have a product with ten components, you need ten fixtures and ten programs for component testing. Of course, even a manufacturing defect analyser requires a dedicated fixture and program but these two cost very less. For ICT the cost is higher due to full power-on testing.

Next comes the functional testing of the product. A functional tester (FT) ensures that the product is work-

ing as per the requirements. There are various kinds of functional test equipment. Sometimes you may use a million-dollar machine and sometimes not even a million-rupee machine. It goes like this: If your product is very simple, you can develop a smaller fixture and some back-end signalling for your product. For example, electronic energy meter manufacturers need to check whether the LED displays properly or not when electricity is fed, etc. They want to see the sequence of their product in powering condition as well as in functioning condition. For this, a very small FT which is very cheap can be developed. But if the product is critical, the cost of an FT goes very high.

Functional testers are dedicated testers. That means if you develop a fixture for an energy meter, it can be used for a particular energy meter or board only. This functional testing is cheaper. But if you want a universal functional tester, it will be a huge machine that can test a hundred types of boards. This machine is very costly.

MDAs, also called ICTs (but not power-on ICTs), cost \$25,000-35,000. The machine's cost is based on its pin configuration. If the board has 500 points, an ICT with 500 outputs is required.

ICTs cost minimum \$100,000-150,000. For a very good brand, the cost can go up to \$300,000. Universal FTs cost million dollars or more.

Automatic optical inspection (AOI) systems (used to detect manufacturing defects) do optical testing only. Sometimes designers in the manufacturing line try to inspect the assembly for component defect with naked eye. AOI system helps to ensure that faults aren't skipped. It can be desktop or inline type. Desktop AOI system costs \$40,000-50,000, while inline AOI system costs \$70,000-90,000.

Another machine which the manufacturer may want to use is the flying probe tester (FPT). Its speed of testing is very slow but no fixture is needed—fixture adds a lot of cost. But it needs a program, which is very easy to develop. FPT costs about \$150,000.

X-ray testers are very dedicated for ball-grid array (BGA) components. Because an AOI system can't inspect this particular type of device, a few manufacturers who want 100 per cent testing—as required in critical applications like avionics, automotive electronics and space electronics—use automated X-ray inspection (AXI) testers also. Automated X-ray testers cost \$100,000-200,000.

What adds to the cost of manufacturing test?

Testing, once confined exclusively to the end of the line, now occurs at several points to provide greater failure isolation and feedback to the manufacturing process. Process tests such as ICT, AXI and AOI sit at various points along the line to immediately catch process flaws. Functional test, which tests the operation of the system of components, typically resides at one or more points along the line and at the end of the line as the final test of the fully assembled product.

With decreasing component size and correspondingly increasing circuit board density and complexity, many products no longer have access points for the bed-of-nails fixtures used for in-circuit testing. Although AOI and AXI systems continue to become more common and more comprehensive, functional test is used more extensively to find and even isolate faults.

With the need for higher throughput and more efficient production and testing, contract electronics manufacturers (CEMs) and OEMs demand sophisticated test platforms for functional testers that are built on open industry standards.

New approaches to testing

There have been efforts to make manufacturing test cost-effective, simpler and more efficient.

Naresh Kumar, general manager-South Asia Pacific Application Engineering Organization, Agilent Technologies, explains that the cost of a material to be tested is amortised over